RAG Assignment: Baseline vs. Re-Ranking Report

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GitHub Repository: `https://github.com/gokulc01/two-stage-rag-pipeline`

1 Architecture Comparison

Bi-encoders let you calculate vectors beforehand for your queries and documents so searching is fast. Retrieval just checks the similarity, making it highly efficient for searching large databases.

But, a cross-encoder requires running both the query and the document simultaneously through the model at runtime. Therefore model capture detailed token interactions between the query and the text, giving you highly accurate relevance scores, but it is slower due to added computation.

2 Qualitative & Rank Shift Analysis

I used the FIDE Laws of Chess rulebook for this assignment. I tested two queries where the baseline Bi-Encoder struggled with complex exceptions, but the Cross-Encoder successfully moved the critical information into the top 3 chunks.

2.1 Failure Mode 1: Exception Override

Query: *Under what circumstance does an opponent receive half a point rather than a full victory when an electronic communication device penalty occurs?*

- **Rank Shift:** The critical passage (Chunk 50) sat at Rank #5 in the baseline retrieval. The Cross-Encoder bumped it to Rank #1 (a shift of +4).
- **LLM Answer 1 (Baseline):** "I do not have enough information." The Bi-Encoder retrieved general rules but missed the specific exception clause.
- **LLM Answer 2 (Re-Ranked):** "If the opponent cannot win the game by any series of legal moves, his score shall be a draw." With the correct chunk provided, the LLM successfully identified the exact corner case condition.

2.2 Failure Mode 2: Unsupervised Venue Procedure Override

Query: *If no arbiter is present in the venue during a quickplay finish, what specific documentation must a player submit to claim a draw on the basis that their opponent is making no effort to win?*

- **Rank Shift:** Chunk 73 moved from Rank #6 to Rank #1 (a shift of +5).
- **LLM Answer 1 (Baseline):** "I do not have enough information." The baseline results filled the context window with general arbiter duties (Chunk 49, Rank 2).

- **LLM Answer 2 (Re-Ranked):** "The player must write down the final position and submit an up to date scoresheet." Re-ranking surfaced Appendix D.1, ensuring the LLM had the specific rules required when no arbiter is present.

3 Latency Profiling

I measured the computation time of re-ranking against the baseline retrieval. While retrieval operations take milliseconds, LLM inference takes most of the total pipeline duration.

Query Index	**Baseline Retr.**	**Re-Rank Time**	**Pipe 1 Total**	**Pipe 2 Total**
Query 1 (Device Penalty)	22.34 ms	53.80 ms	11.319 s	8.814 s
Query 2 (No Arbiter)	8.98 ms	55.97 ms	11.371 s	9.838 s

Table 1: Latency comparison between Baseline and Re-Ranking pipelines.

Note: The total end-to-end time varied largely due to LLM generation speeds for different response lengths.

4 Scaling Question

Because cross-encoders need both the query and the text at the same time to calculate relevance, so no pre-calculations. If you had 1,000 documents, a cross-encoder would have to run 1,000 separate times for a single query, which would take way too long. The two-stage pipeline solves this by using the fast bi-encoder to narrow the field to a small subset (like the top 5), and then using the cross-encoder just for those few to get the final ranking.